Look and Pick: Gaze-Directed Target Selection for a Robot Arm with Smart Glasses

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Motivation

Telling a robot *which* object you want is still awkward: joysticks, screens, or naming it.

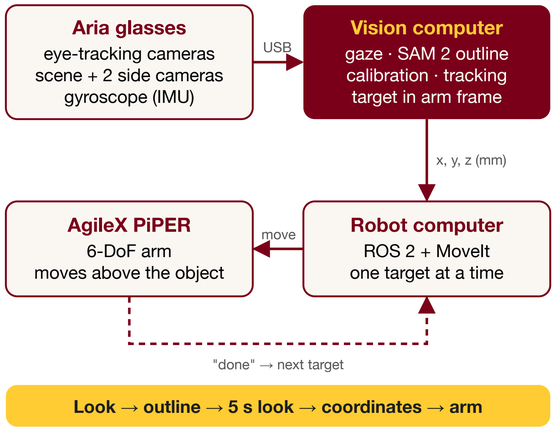
People already point with their eyes. For someone with limited hand use, a robot that understands "that one" from a glance would make everyday manipulation far more natural.

Objectives

1. **Choose with the eyes:** select any object by looking at it, with no object list or training.
2. **Locate it for the robot:** find its exact position in the arm's own coordinates, without markers.
3. **Deliver it safely:** send only reachable targets to the arm, one at a time.

What's new: the robot arm itself is the calibration target, SAM 2 outlines any object, and the camera rides on the wearer's moving head.

System Overview



Selection Pipeline

1. The eye model finds where the wearer looks.
2. SAM 2 outlines **whatever** is under the gaze.
3. It lights up; a steady **5 s look** chooses it.
4. Its position comes out in the arm's frame.

Arm-Based Camera Calibration

The robot is its own ruler. The gripper visits **6 known positions** and its tip is clicked in the frozen frame; matched to the arm's coordinates, the clicks give the camera's pose in the arm's frame (Perspective-n-Point). A **7th point is held out** to measure the accuracy.

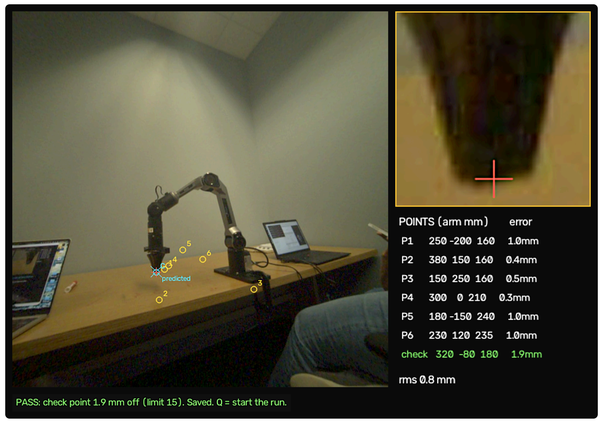


Fig. 1. Calibration screen at PASS: clicked points 1–6 (yellow), the predicted check point (cyan), a close-up of the gripper tip, and the error at each point.

- Every point is checked against the PiPER's joint model before use.
- A bad click is detected and redone on its own.
- If the wearer moves, it says which way to move back.

Gaze-Guided Segmentation

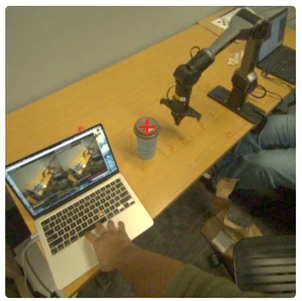


Fig. 2. Live view from the glasses; the red cross is the gaze.

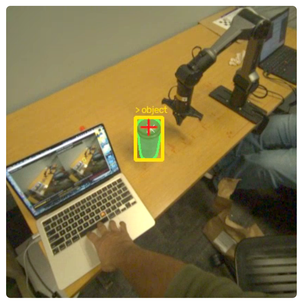


Fig. 3. SAM 2 outlines the cup under the gaze, with no object list.

An object lights up after a fifth of a second of steady looking; looking just beside it snaps to the nearest one. Holding the look for **5 s** chooses it.

Multi-Target Selection

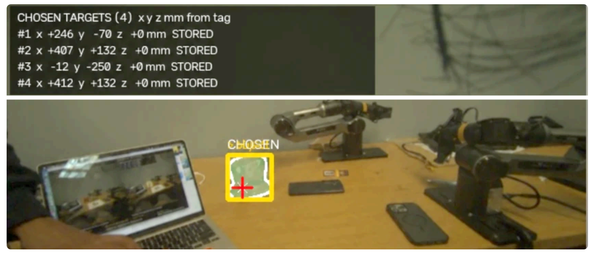


Fig. 4. Objects are chosen one after another and stored in a list (top). The arm gets only the first; the next is sent after the arm replies *done*. Earlier run, when a printed tag still gave the arm's frame.

Robot Platform and Control

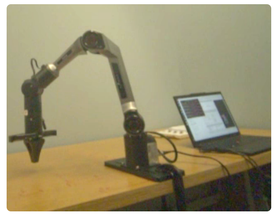


Fig. 5. AgileX PiPER, gripper pointing down.

Joints	**6**
Reach	**626 mm**
Payload	**1.5 kg**
Repeatability	**± 0.1 mm**
Gripper	**0–70 mm**
Control	**ROS 2 + MoveIt**

Each target arrives as one message (x, y, z in mm, arm frame). The planner solves the joint angles, plans a smooth path, and moves with the gripper pointing down, first to a point **above** the object. A table model checks the whole path. When a move ends, the robot computer replies *done* and the next chosen object is released.

Results

0.8–1.1 mm
calibration error (RMS), 6 arm points
2 hardware calibrations · per point 0.2–1.6 mm

1.9–3.0 mm
held-out check point error
never used in the solve · pass limit 15 mm

≤ 1 mm
arm reaching its commanded points
6 calibration points · 2 sessions

0.3 s
object outline (SAM 2), single GPU
full calibration ≈ 4 min, no markers

Reachable Workspace

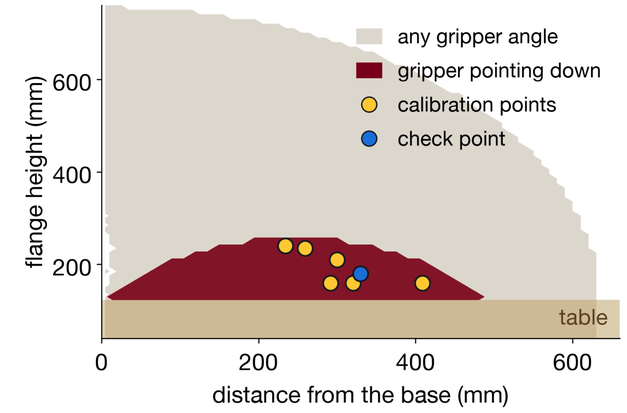


Fig. 6. From the PiPER's published joint model. Purple: reachable with the gripper pointing down. All 7 calibration points lie inside it.

Visual-Inertial Head Tracking

After calibration the wearer may move. Table features in the two side cameras and the scene camera are matched to the calibration moment, and the **gyroscope (1 kHz)** carries the head's rotation through blur and brief loss of view. When tracking is uncertain, nothing is selected or sent.

Simulated room: median 6 mm, 0.4°; a 1.5 s camera dropout bridged within 1°.

Future Work

- **Grasping:** top-down grasp from the outline: approach from above, align to the narrow side, close, lift.
- **Aria Gen 2:** on-device head tracking for free head movement.
- **Learned grasping (VLA):** a vision-language-action model to pick up uneven or awkward objects that cannot be held from the top.
- End-to-end accuracy and a small user study for assistive use.

Acknowledgements

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